

ESG controversies and corporate credit risk

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Abstract

This thesis examines whether ESG controversies are priced in the credit-risk market. It uses a panel of 98 non-financial firms from 2010 to 2017, matching RepRisk controversy data with credit default swap (CDS) spreads and firm-level controls. It tests whether controversy incidents, media reach, and severity are associated with CDS spreads. A level specification returns a negative coefficient on incident count. This reflects measurement rather than underlying risk, since RepRisk's media coverage expanded during a period of improving credit quality. Controlling for the lagged CDS spread removes this confound, and the coefficient falls to zero. The reach effect survives this correction, and spreads widen following high-reach incidents. Severity shows no comparable effect. The results show that incident counts can carry the wrong sign unless specifications isolate within-firm repricing from persistent media-attention differences, and that the reach mechanism documented by Kölbel et al. (2017) holds in a later sample period.

Keywords: ESG controversies; credit default swaps; corporate credit risk; media reach; stakeholder sanctions

1. Introduction

The relationship between Environmental, Social, and Governance (ESG) factors and corporate financial outcomes has become one of the most active areas of research in corporate finance, linking a firm's ESG profile to its risk, financial performance, and value. A useful distinction within this literature is between a firm's ongoing ESG performance, which is typically captured by slow-moving ratings, and its ESG controversies. These controversies are specific, dated incidents in which a firm is implicated in environmental, social, or governance misconduct. Unlike ratings, controversies carry precise informational content and occur at an identifiable point in time. This makes them particularly well suited to event-based tests of how financial markets price non-financial risk.

Existing evidence on the pricing of ESG controversies has focused primarily on equity markets and on aggregate measures of the cost of debt. Creditors, however, occupy a distinctive position among a firm's stakeholders. They bear the downside of default risk without sharing symmetrically in the upside of firm performance. This asymmetry, consistent with the credit risk framework of Merton (1974), makes creditors naturally more sensitive to bad news than equity investors are. Once the probability of default rises, creditors are therefore likely to react sharply. The credit default swap (CDS) market offers a direct, market-based measure of this reaction.

Two studies provide the closest precedent for this thesis. Kölbel et al. (2017) study an international sample of 539 firms between 2008 and 2013. They find that media coverage of corporate social irresponsibility is a substantial driver of financial risk. This effect depends heavily on the reach of the coverage and, conditional on reach, on its severity. Barth et al. (2022) study a broader panel of European and US firms and show that higher aggregate ESG ratings are associated with lower CDS spreads. This thesis does not identify a previously unstudied relationship. It re-examines the reach and severity mechanisms identified by Kölbel et al. (2017), separating the level relationship between controversy exposure and CDS spreads from the within-firm repricing that occurs at the time of an event. This distinction is not tested in the original study. The analysis also draws on a more recent sample period (2010–2017).

This thesis uses a panel of 98 non-financial firms observed quarterly between 2010 and 2017. It combines event-level ESG controversy data from RepRisk with CDS spreads and firm-level accounting and ESG controls to test three hypotheses. The first

examines whether the occurrence of ESG controversy events is associated with higher CDS spreads. The second examines whether controversies reported by higher-reach media sources show a stronger positive association with CDS spreads. The third examines whether the severity of the criticism further amplifies this relationship, conditional on high media reach. These hypotheses are tested using a level specification, which is consistent with the baseline approach in the literature. They are also tested using change and dynamic specifications, which are intended to separate genuine within-firm repricing from persistent, unrelated differences in media exposure across firms.

The results do not support a simple positive relationship between the number of controversy incidents and CDS spreads. In the level specification, the coefficient on incident count is negative and significant. This does not mean that controversies lower credit risk. RepRisk expanded its media coverage over the sample period, so later incidents partly reflect closer monitoring rather than a genuine rise in misconduct. This period also coincides with improving credit quality for many of the firms in the sample. Once the lagged CDS spread is added to account for this persistence, the coefficient on incident count becomes statistically indistinguishable from zero. The reach of media coverage, by contrast, remains associated with a significant widening of spreads, an effect concentrated in high-reach incidents specifically. The severity of the underlying misconduct does not significantly moderate this relationship. These patterns are broadly consistent with the mechanism documented by Kölbel et al. (2017), with reach again emerging as the more robust dimension.

The contribution of this thesis is twofold. Methodologically, it shows that the sign and significance of the count-based exposure measure depend materially on whether the specification isolates within-firm repricing from persistent cross-sectional differences in media exposure. Empirically, it provides an out-of-sample replication of the reach-driven mechanism identified by Kölbel et al. (2017), confirming that the effect survives in a more recent sample period and remains concentrated in high-reach incidents specifically.

The remainder of the thesis proceeds as follows. Section 2 reviews the relevant literature and develops the three hypotheses. Section 3 describes the data and the empirical methodology. Section 4 presents the empirical results, from the baseline specification through the robustness tests. Section 5 concludes.

2. Literature Review and Hypotheses Development

2.1 ESG Performance, Controversies and Firm Risk

Whether a firm's Environmental, Social, and Governance (ESG) performance affects its financial outcomes has become a central question in corporate finance, particularly for firm risk and value. Gillan et al. (2021) provide a comprehensive review of this literature, showing that a firm's ESG profile is strongly related to its corporate risk, financial performance, and overall firm value. ESG indicators have therefore moved beyond their ethical origins to become a core element of the corporate finance environment.

Using a global sample of firms, Lee and Faff (2009) find that companies with leading sustainability performance exhibit significantly lower idiosyncratic risk than their lagging counterparts. This lower risk also appears to be priced by the global equity market. A parallel pattern appears in the corporate social responsibility (CSR) literature, where Godfrey et al. (2009) similarly link responsible corporate conduct to lower firm risk following negative events.

However, this risk-reducing relationship reverses when firms neglect their ESG responsibilities and become involved in negative incidents. Shakil (2021) shows that while good ESG performance generally reduces total firm risk, involvement in ESG controversies plays a moderating role. Controversies weaken the risk-mitigating effect of ESG performance and directly raise overall financial risk. This asymmetry indicates that ESG misconduct does not simply reverse the benefits of positive ESG performance. Its financial consequences therefore deserve attention as a research question of their own.

The evidence reviewed so far concerns firm-level total risk and overall financial performance. These outcomes matter to both equity holders and creditors, but through different channels. Equity holders bear the firm's residual risk, while creditors are principally concerned with the probability that a firm will fail to meet its debt obligations. ESG controversies raise the volatility of expected cash flows and increase the likelihood of adverse legal, regulatory, and reputational shocks. In principle, these risks should therefore be reflected in the price of both debt and equity capital. The following sections examine this question directly, first reviewing the literature on how ESG risk is priced in debt and equity markets, and then turning to credit default swap

spreads as a measure of corporate credit risk.

2.2 The Pricing of ESG Risk in Debt and Equity Markets

Several strands of research examine how ESG-related risks are priced in corporate debt and equity markets. Because such risks raise the required risk premium demanded by capital providers, they should be reflected in the cost of both debt and equity financing. Early evidence on this relationship comes from studies examining the environmental dimension of ESG risk. Chava (2014) demonstrates that firms with significant environmental concerns face materially higher costs of both equity and debt capital. Lenders charge higher interest rates on bank loans to environmentally sensitive firms, and fewer banks participate in their loan syndicates, reflecting a reduced willingness to extend credit. Importantly, this premium reflects new information about firm risk that standard financial measures do not already capture. Comparable evidence emerges on the equity side. El Ghoul et al. (2011) find that firms with stronger ESG performance exhibit a significantly lower cost of equity capital, with responsible employee relations, environmental policies, and product strategies contributing most to the reduction. Because this thesis focuses on corporate credit risk, the remainder of this review turns to how creditors specifically respond to ESG risk.

This question of how creditors respond is taken up directly by Oikonomou et al. (2014), who examine multiple dimensions of corporate social performance. Using a longitudinal sample of corporate bond issuances, they find that strong social and environmental performance is rewarded with lower bond yield spreads, while social transgressions lead to wider spreads and lower credit ratings. Their results also reveal an asymmetry, with the penalty for negative ESG behavior more pronounced than the reward for positive behavior. This fits what theory predicts. Creditors bear the full loss when a firm performs badly but gain nothing extra when it performs well, so they weight bad outcomes more heavily when pricing debt. It also suggests that negative ESG events deserve particular attention, since their effect on credit risk may exceed what the broader ESG-performance literature implies.

The same pattern extends from public bond markets to private credit, where banks act as specialized monitors of corporate risk. Goss and Roberts (2011) analyze a large sample of US bank loans and find that firms with social responsibility concerns pay between 7 and 18 basis points more in loan spreads than more responsible

borrowers. This premium is largest when a loan has no collateral, when lenders have less to fall back on and so rely more heavily on non-financial signals. Combined with the bond-market evidence, these findings show that debt markets systematically incorporate ESG-related information into credit pricing. However, this evidence relies largely on aggregate measures of social performance rather than on discrete negative events.

A more recent strand of research turns directly to discrete ESG controversy events rather than accumulated ratings. Du and Lu (2025) investigate how ESG incidents affect corporate debt financing costs, using RepRisk incident data matched with Chinese listed firms. They find that ESG incidents raise debt costs by approximately 70 basis points, an effect that operates through four channels. The incidents elevate credit risk, damage banking relationships, weaken information quality, and depress operating performance. Their pillar-level analysis further reveals that social and governance incidents are the primary drivers of increased debt costs, whereas environmental incidents show no significant effect. This pattern suggests that creditors respond differently to different categories of ESG misconduct.

Arcidiacono et al. (2026) complement this evidence from a European perspective, examining how ESG controversies affect the cost of both debt and equity for firms in the STOXX Europe 600. Their findings confirm that ESG controversies significantly increase financing costs in both markets, with effects that vary across ESG sub-pillars. Consistent with Du and Lu (2025), they find that social controversies are the dominant driver of higher debt costs, whereas equity investors respond more uniformly across all types of controversy. Their mediation analysis further identifies market uncertainty as a key transmission channel. ESG controversies erode a firm's perceived legitimacy and heighten uncertainty about its future cash flows and regulatory exposure. This rising uncertainty, in turn, raises the risk premia demanded by both lenders and shareholders.

Beyond the occurrence of an incident, the way it becomes public also shapes its financial impact. Kölbel et al. (2017) show that media coverage substantially mediates the financial risk arising from corporate misconduct. Using RepRisk data on an international panel of firms, they find that media coverage of corporate social irresponsibility significantly increases credit risk, an effect driven primarily by articles published in high-reach global outlets.

Overall, these studies demonstrate that ESG-related risks are consistently priced in both debt and equity markets, across different institutional contexts and measurement approaches. What remains less settled is how credit risk itself should be measured. The following section turns to this question, examining the limited evidence on ESG and CDS markets and identifying the research gap that the present study addresses.

2.3 Measuring Credit Risk with CDS Spreads

The preceding sections establish that ESG-related risks are priced in both debt and equity markets, but how credit risk itself should be measured is a separate question. Most existing studies measure the cost of debt either with accounting figures, such as a firm's interest expense divided by its total debt, or with bond yield spreads. Accounting-based debt costs are low-frequency and backward-looking, reflecting historical borrowing conditions rather than current market assessments. Bond yield spreads come from the market, but are affected by economy-wide interest-rate changes and by tax rules that differ across countries, making it hard to isolate the part of credit risk that is specific to the individual firm (Kölbel et al., 2017).

Credit default swap spreads offer a superior alternative. A CDS spread, expressed in basis points, represents the annual cost of insuring against a firm's default, and so gives a direct, market-based measure of the market's assessed probability of default. This fits the structural view of credit risk, in which a firm defaults when the value of its assets falls short of what it owes (Merton, 1974). CDS spreads are also less distorted than bond yields by liquidity premiums, coupon structures, or embedded optionality, and the CDS market is highly liquid and updates continuously, allowing researchers to capture real-time shifts in market sentiment around specific events (Kölbel et al., 2017). The five-year CDS spread, the most liquid and most accurately priced segment of the market, has consequently become the standard instrument for measuring corporate credit risk (Barth et al., 2022).

Beyond these measurement advantages, the institutional design of the CDS market gives it a distinctive economic role in revealing credit deterioration, which is particularly relevant to the study of discrete negative events. A CDS is an unfunded derivative, so an investor who is pessimistic about a firm's credit can act on that view directly by buying protection. This is far simpler than borrowing and short-selling the underlying corporate bond, which is often hard to trade (Augustin et al., 2014). The

CDS market therefore offers a lower-friction venue for expressing negative credit views than the cash bond market. Protection buyers are typically motivated by the desire to hedge tail risk, the low-probability but high-loss outcomes an ESG controversy can suddenly bring forward. This gives CDS spreads an asymmetric, downside-oriented character, making them especially responsive to adverse shocks. Consistent with this view, the empirical literature documents that the CDS market frequently leads the bond market in price discovery, incorporating new firm-specific credit information before it is reflected in bond yields (Blanco et al., 2005). This informational advantage is most pronounced when the news is bad. Acharya and Johnson (2007) find that information flows from the CDS market to the equity market are concentrated in negative-news events and in firms that subsequently experience credit deterioration. In combination, these features imply that if ESG controversies carry genuine credit-relevant information, the CDS market is the venue in which that information should surface earliest and most cleanly. This rationale for studying CDS spreads over bond yield spreads is therefore economic as well as statistical.

Few studies have examined the relationship between ESG factors and CDS spreads directly. Barth et al. (2022) provide the first systematic investigation of whether the CDS market reflects ESG performance, comparing firms in Europe and the United States. They use a comprehensive panel dataset spanning 2007 to 2019. They find that higher ESG ratings significantly reduce CDS spreads. A one-standard-deviation improvement in ESG rating lowers spreads by approximately 4% to 8%, depending on the firm's ESG quantile. Their results further identify a mediating volatility channel. Strong ESG performance lowers firm-level earnings volatility, which in turn reduces the credit risk priced in the CDS market. This indirect channel amplifies the direct effect of ESG on CDS spreads, indicating that the relationship between non-financial performance and credit risk is more complex than linear models would imply.

Existing research on ESG and CDS spreads has focused mostly on aggregate ESG ratings, which reflect a firm's overall sustainability profile accumulated over time. This differs from examining discrete ESG controversy events, which represent sudden negative shocks to a firm's perceived legitimacy and risk profile. Such events carry clear informational content and occur at identifiable points in time, and they have been shown to trigger rapid revaluations of corporate risk in both equity and debt markets (Du and Lu, 2025; Arcidiacono et al., 2026). Kölbel et al. (2017) provide direct

evidence that such events also move CDS spreads, and that this reaction is stronger when a controversy receives wider media coverage.

The stakeholder sanction mechanism identified by Kölbel et al. (2017) offers a useful starting point for this study. It implies that a controversy's effect on credit risk should depend not only on whether it occurs, but also on how visible and how serious it is. Media attention coordinates stakeholder responses, which in turn raises the expected cost of default and widens CDS spreads. Testing this reasoning empirically requires event-level ESG data of the kind provided by RepRisk, which records an incident's occurrence together with its reach and severity, data that have only recently become available to researchers. Using this data, the present study examines how exposure to ESG controversy incidents, and the reach and severity with which they are reported, are associated with corporate credit risk.

2.4 Hypotheses Development

The previous sections suggest a clear way in which ESG controversies can affect a firm's credit risk. A natural starting point is the risk-management view of CSR by Godfrey et al. (2009). They argue that acting responsibly builds up goodwill, or "moral capital," among a firm's stakeholders. This goodwill works like insurance. When something bad happens, it softens how harshly stakeholders punish the firm. An ESG controversy can be seen as using up or even destroying this protective cushion. However, this insurance idea mainly explains why a firm with goodwill is partly protected. It does not directly explain why a controversy makes a firm look riskier to its creditors. Two further mechanisms help make this clear.

First, a controversy can directly hurt the firm's cash flows and the value of its assets. It often leads to lawsuits, fines, and clean-up costs. It can also reduce future income through consumer boycotts, lost contracts, or less motivated employees. Under the structural view of credit risk introduced earlier (Merton, 1974), a lower and more uncertain asset value pushes a firm closer to default. It also reduces what creditors could recover if the firm fails. Creditors do not benefit when a firm does well, so a rise in expected losses like these weighs heavily on how they judge default risk.

Second, a controversy also sends a signal about information the firm may have been hiding. It can reveal weaknesses in the firm's risk management, internal controls, or governance that outsiders did not know about, and it raises the worry that more bad

news has not yet come out. Cui et al. (2018) find that firms which behave more responsibly tend to have lower information asymmetry with outside investors. A controversy works in the opposite direction. It widens the information gap between managers and outside investors. Because creditors dislike this added uncertainty, they demand a higher return to compensate for it.

Both of these mechanisms depend on how stakeholders react, an idea at the heart of stakeholder theory (Freeman, 1984) and developed further by Kölbel et al. (2017). Once a controversy becomes widely known, stakeholders tend to act against the firm together, through regulatory scrutiny, lawsuits, and damage to its reputation. This turns general disapproval into real financial losses. In this picture, the moral capital stressed by Godfrey et al. (2009) mainly decides how hard these reactions hit the firm. The reactions themselves, and the extra uncertainty they create, are what actually push creditors to see the firm as riskier.

So far, most studies have shown this effect using accounting-based measures of debt cost or bond yield spreads (Goss & Roberts, 2011; Oikonomou et al., 2014; Du & Lu, 2025; Arcidiacono et al., 2026). Extending it to CDS spreads is a natural next step. The CDS spread is the most direct market-based signal of default risk, so it should pick up this kind of information at least as quickly as those slower measures. When the market raises its view of a firm's default risk, the CDS spread widens. Based on this reasoning, this paper proposes the following hypothesis:

H1: Greater exposure to ESG controversy incidents is positively associated with CDS spreads.

Not all controversy incidents are equally visible to market participants. RepRisk classifies the reach of each information source according to the potential audience and influence of the source. This distinction matters because a controversy reported by a high-reach source is more likely to be observed by creditors, investors, regulators, customers, and other stakeholders. Wider visibility can coordinate stakeholder attention and increase the likelihood of sanctions, reputational losses, and market repricing.

Kölbel et al. (2017) show that the effect of CSI coverage on credit risk is driven especially by high-reach media coverage. More recent RepRisk-based evidence supports the relevance of this mechanism. Gutknecht (2025) finds that CSI coverage increases several dimensions of firm equity risk and that the effect is strongest for high-reach media sources. Mendiratta et al. (2023) similarly show that ESG controversies

reduce firm value in India mainly when media reach is high. Akyildirim et al. (2025) also treat reach as an important characteristic of ESG reputational events when analyzing market responses around Brexit. Collectively, these studies suggest that the market impact of ESG controversies depends on both how many incidents occur and how widely they are reported.

H2: ESG controversy incidents reported by higher-reach sources have a stronger positive association with CDS spreads.

Besides reach, the severity of criticism also shapes market reactions. The empirical evidence on severity is less uniform than the evidence on reach, but it remains theoretically relevant. Kölbel et al. (2017) find that severity does not have a clear standalone effect in their main model, but that it reinforces the effect of coverage when media reach is high. Mendiratta et al. (2023) also examine severity as a core media channel through which ESG controversies affect firm value. Gutknecht (2025), however, finds that severity does not consistently moderate the relationship between CSI coverage and equity risk. These mixed findings make severity an important empirical question rather than a settled assumption. If more severe ESG controversies convey stronger information about potential stakeholder sanctions and future cash-flow losses, they should be more strongly associated with CDS spreads.

H3: ESG controversy incidents with higher severity have a stronger positive association with CDS spreads.

Taken together, these three hypotheses form the core empirical framework of this study. H1 establishes the baseline relationship between exposure to ESG controversies and credit risk as priced in the CDS market. H2 and H3 then identify the conditions under which this relationship is amplified. The subsequent empirical analysis tests these hypotheses using panel data on ESG controversy events from RepRisk, matched with CDS spread data and firm-level financial controls.

3. Data and Methodology

This study examines whether ESG controversies are associated with corporate credit risk. Credit risk is measured using credit default swap (CDS) spreads, while ESG controversy exposure is measured using the RepRisk Historical Risk Incidents and Historical Metrics files. The empirical setting is a quarterly firm-level panel covering the period from the first quarter of 2010 to the fourth quarter of 2017.

3.1 Data Sources

3.1.1 Financial and Accounting Data

Financial and accounting data are obtained from LSEG Workspace. This source provides the daily CDS quotes used to construct the dependent variable, together with the annual accounting variables, industry classifications, and credit-rating categories used in the analysis. An annual firm-level ESG score is also obtained from the same source and is used as a control variable. The analysis uses the five-year CDS contract, which is the most liquid and therefore the most accurately priced tenor. The initial dataset contains daily observations for 163 corporate reference entities.

3.1.2 ESG Controversy Data

ESG controversy information is obtained from RepRisk. RepRisk screens media, stakeholder, and other third-party sources on a daily basis and identifies news items that criticise companies for environmental, social, and governance misconduct. RepRisk does not assess whether an accusation is true, so the data record what external stakeholders report rather than verified misconduct. This feature is appropriate for the present setting, because credit spreads respond to information that reaches the market rather than to allegations that have been legally established.

Two complementary RepRisk datasets are used: Historical Risk Incidents and Historical Metrics. The Historical Risk Incidents files provide event-level information. Each observation identifies a RepRisk company, a story, an incident date, whether the incident is classified as unsharp, the severity of the incident, media reach, novelty, and environmental, social, governance, and cross-cutting classifications. The Historical Metrics files provide daily company-level risk indicators. These indicators aggregate the underlying incidents into a time-varying measure of reputational risk exposure.

3.2 Sample Construction

The sample is restricted to non-financial firms because financial firms differ substantially from non-financial firms in their leverage, liquidity, balance-sheet structure, regulation, and interpretation of accounting ratios. In particular, the current ratio and debt-to-equity ratio are not directly comparable between banks, insurers, and industrial firms. The exclusion of financial firms is common in studies of the relation between corporate social performance and the cost of capital (Chava, 2014; Cheng et

al., 2014).

A firm-quarter is retained only if the CDS spread is observed and if the ESG score and all accounting controls are available for the preceding fiscal year. After merging the CDS data with annual ESG scores, accounting variables, industry information, and credit-rating categories, the non-financial complete-case sample contains 105 firms and 2,921 firm-quarter observations.

RepRisk companies are linked to the CDS sample through ISINs rather than company names, since names differ across providers and could produce incorrect matches. Of the 105 firms, 98 are uniquely matched to a RepRisk Company ID. The remaining seven are excluded rather than coded as having zero controversies, since a failed match reflects incomplete identifier coverage.

The incident-level records are processed before aggregation. The annual source files covering 2010 to 2017 are pooled into a single dataset, and duplicate events are removed using the combination of RepRisk Company ID, Story ID, and Incident Date, ensuring that a story is not counted more than once. Each incident is assigned to a calendar quarter using its incident date and aggregated to the firm-quarter level for the 98 matched firms. The metrics files are aggregated to the firm-quarter level in the same way.

The final estimation sample contains 98 non-financial firms and 2,737 firm-quarter observations. This is close to, but below, the theoretical maximum of 3,136 observations (98 firms across 32 quarters from 2010Q1 to 2017Q4). The panel is therefore unbalanced: 71 firms have the full 32 quarters, while 27 have fewer, which gives a median of 32 quarters per firm but a mean of only 27.9. The shortfall reflects missing CDS quotes or missing lagged accounting and ESG data in the affected firm-quarters.

3.3 Variable Definitions

3.3.1 Dependent Variable

The dependent variable is the corporate CDS spread. To align the dependent variable with annual accounting data, annual ESG scores, and quarterly RepRisk measures, the analysis is conducted at the quarterly frequency. The baseline CDS measure is the last available positive CDS quote for each firm within each calendar quarter. This variable is denoted as *cds_eoq*. In practice, the observation is the final

available trading-day quote in the quarter.

Quarter-end CDS spreads are used as the main outcome for three reasons. First, they provide a clearly defined market valuation date for each firm-quarter. Second, they avoid giving disproportionate weight to quarters with more daily observations. Third, and most importantly for the research question, they ensure that the dependent variable is measured after the controversy incidents recorded during the same quarter, which fixes the direction of the timing between the explanatory and the dependent variable. Because a single end-of-quarter quote can be affected by short-lived liquidity conditions or idiosyncratic market noise, a quarterly-average CDS spread is also constructed as a robustness measure. This variable, *cds_quarter_average*, is calculated as the mean of all available daily positive CDS quotes within a firm-quarter. It is not used as the baseline because part of it is measured before the incidents recorded in the same quarter.

The main dependent variable is the natural logarithm of the quarter-end CDS spread. The logarithmic transformation reduces the influence of the strongly right-skewed distribution of CDS spreads and allows coefficients to be interpreted approximately as percentage changes in credit spreads. The logarithm of the quarterly-average CDS spread is used in an alternative specification. A further specification uses the quarterly change in the logarithm of CDS spreads, defined as the log spread in quarter q minus the log spread in quarter $q-1$. This change specification focuses on short-run CDS repricing rather than on cross-sectional differences in the level of credit risk. Both specifications are reported because they answer different questions. Recorded incident counts capture both adverse events and the intensity with which a firm is monitored by the media, and the second component is persistent within firms.

3.3.2 Independent Variables: ESG Controversy Exposure

The main controversy measure is the number of sharply matched incidents in a firm-quarter, *IncidentsSharp*. An incident is classified as sharp when the RepRisk indicator Unsharp Incident equals zero. This is the preferred measure because sharp incidents represent more direct and precise links between the firm and the reported ESG controversy. The total number of incidents, including unsharp incidents, is retained as an alternative measure, *IncidentsAll*. For robustness, logarithmic forms $\ln(1+IncidentsSharp)$ and $\ln(1+IncidentsAll)$ are constructed. Adding one allows firm-

quarters with no incidents to remain in the sample.

The incident data also permit tests of the characteristics of controversy exposure. Media reach is classified into low, medium, and high categories. Accordingly, the study constructs quarterly counts of low-, medium-, and high-reach incidents. The underlying idea is that controversies reported by sources with wider reach are more likely to be observed by creditors, investors, regulators, customers, and other stakeholders. If visibility strengthens market discipline or expected stakeholder sanctions, high-reach incidents should have a larger positive association with CDS spreads. This mechanism is supported by recent evidence that the market response to ESG controversies is concentrated in incidents that receive wide media dissemination (Mendiratta et al., 2023; Gutknecht, 2025).

Similarly, incident severity is classified into low, medium, and high categories. Quarterly counts are constructed for each severity level. More severe incidents may imply larger expected legal costs, regulatory penalties, operational disruption, reputational damage, or cash-flow losses. The analysis therefore tests whether high-severity incidents have a stronger association with CDS spreads than low-severity incidents. The reach and severity counts are constructed from all recorded incidents rather than from sharply matched incidents only, since the classification applies to every incident in the source files. Because the three reach counts and the three severity counts each sum to the total number of incidents, they are never included in the same specification as *IncidentsSharp* or *IncidentsAll*.

In addition to these counts, the average reach and the average severity of the incidents recorded in a firm-quarter are constructed on the same three-point scale. These two variables are estimated jointly with the incident count, so that they capture the intensity of controversy exposure conditional on its volume. Their interaction is included in a further specification, which tests whether severe criticism has a stronger effect when it is also widely disseminated. This design separates the number of controversies from their characteristics, which is necessary because a firm may accumulate many minor incidents without attracting broad attention.

Firm-quarters in which no incident is recorded take the value zero for all incident counts and for average reach and average severity. These observations are not missing values but genuine cases of zero exposure, and they are retained in the sample. The Historical Metrics files provide an alternative to event counts. The analysis uses

the Current RepRisk Index on the last available day of each quarter. The Current RepRisk Index weights incidents by reach, severity, and novelty and decays over time in the absence of new criticism, so it measures the accumulated stock of reputational risk exposure rather than the flow of new events. It is the main metrics-based alternative to the incident counts. It is never estimated in the same model as the incident counts, because it is itself constructed from the underlying controversy events.

3.3.3 Control Variables

The regressions include an annual ESG score as a measure of the broader ESG performance of the firm. The score is lagged by one fiscal year, so that CDS spreads in year t are matched with the score for fiscal year $t-1$. This ensures that the score is publicly available before the credit risk outcome is measured and reduces simultaneity concerns. The same timing convention is applied in studies relating corporate social performance to the cost of debt (Oikonomou et al., 2014). Including the score also separates the effect of realised controversies from the effect of the general ESG profile of the firm, which prior work finds to be negatively related to financing costs (El Ghouli et al., 2011; Cheng et al., 2014).

The annual accounting controls are also lagged by one fiscal year and follow the structural determinants of default risk, in which a firm defaults when the value of its assets falls below the value of its obligations (Merton, 1974). The controls are leverage, defined as total debt divided by common equity; the current ratio, defined as current assets divided by current liabilities; the operating profit margin; and firm size, measured as the natural logarithm of net sales, since the raw sales distribution is highly skewed. Leverage, the current ratio, and the operating profit margin are winsorised at the 1st and 99th percentiles in the regression sample to reduce the influence of extreme observations, particularly those driven by very low or negative equity values. Winsorising financial variables at these thresholds is standard practice in this literature (Chava, 2014). Credit quality is captured by a high yield indicator that equals one for firms classified as speculative grade and zero for investment grade firms.

3.4 Methodology

3.4.1 Baseline Specification

The baseline model relates the natural logarithm of the quarter-end CDS spread

of a firm in a given quarter to its ESG controversy exposure in that quarter. The model also includes the ESG score and the accounting controls, both lagged by one fiscal year, together with firm and quarter fixed effects. It is estimated by ordinary least squares on the unbalanced panel of 98 non-financial firms.

Firm fixed effects absorb all characteristics of a firm that do not change over time. These include the business model, the country of incorporation, the industry, the credit-rating category, and persistent differences in the degree to which a firm is monitored by the media. Their inclusion matters in this setting, because firms that are structurally more visible accumulate more recorded incidents without necessarily being riskier. Once firm fixed effects are included, the coefficient on controversy exposure is identified by variation within the same firm over time rather than by differences between firms.

Quarter fixed effects absorb shocks that are common to all firms in a given period, such as movements in risk-free rates, in aggregate credit conditions, and in market risk appetite. Their inclusion is necessary because the general level of CDS spreads declined over the sample period as markets normalised after the financial crisis, while the number of recorded controversies increased as RepRisk expanded its source coverage. Without quarter fixed effects, this divergence in aggregate trends could be mistaken for a firm-level relation between controversies and credit risk.

Standard errors are two-way clustered by firm and quarter. This allows for correlation among observations of the same firm over time and among different firms observed in the same period, both of which are likely in a panel of credit spreads (Petersen, 2009). As a robustness check, the baseline is also re-estimated with standard errors clustered at the industry and quarter level; because the sample contains only eight industry groups, a small number of clusters in one dimension, this alternative is reported for comparability with studies that cluster at the industry level but is not used as the primary specification. Industry classifications are used for the clustering of standard errors and for the robustness analysis rather than as regressors.

3.4.2 Testing the Hypotheses

The hypotheses developed in Section 2 are tested in three steps. The first hypothesis concerns the overall level of controversy exposure. It is tested using *IncidentsSharp*, the number of sharply matched incidents recorded in the firm-quarter.

The second hypothesis concerns the visibility of controversies. It is tested in two ways. The total count is replaced with separate counts of low-, medium-, and high-reach incidents, allowing their effects to differ. Average reach is then added to a model that already controls for the number of incidents, so that the effect of visibility can be distinguished from the effect of the number of incidents.

The third hypothesis concerns the severity of controversies and is tested in the same three ways using the severity classification. A final specification includes average reach, average severity, and their interaction, in order to test whether severe criticism is priced more strongly when it is also widely disseminated.

The distinction between the counts and the averages is deliberate. The counts measure how much criticism of a given type a firm receives, while the averages measure the character of that criticism conditional on its volume. Reporting both prevents the effect of the number of incidents from being confused with the effect of their visibility or seriousness.

The Current RepRisk Index is used in separate specifications as an alternative measure of exposure. Because it aggregates the same underlying events, it is not estimated jointly with the incident counts.

3.4.3 Level and Change Specifications

Two versions of the model are estimated. In the first, the dependent variable is the level of the log CDS spread. This specification examines whether firms with greater controversy exposure carry higher spreads once firm and quarter effects are removed. In the second, the dependent variable is the quarterly change in the log CDS spread, while the controversy variables remain in levels. This specification tests whether the spread of a given firm widens in the quarter in which controversy exposure occurs.

The two questions are related but not equivalent, and they need not yield the same answer. Recorded incident counts capture both adverse events and the intensity with which a firm is monitored, and the second component is persistent within firms. Taking first differences removes this persistent component and therefore isolates short-run repricing.

A third specification adds the lagged log CDS spread to the level model. CDS spreads are highly persistent, and this partial-adjustment form allows the coefficient on controversy exposure to be read as the deviation of the spread from its own path rather

than as a difference in its level. The long-run effect implied by this model is obtained by dividing the coefficient on controversy exposure by one minus the coefficient on the lagged spread. Because the panel contains 32 quarters per firm, the finite-sample bias that arises from combining a lagged dependent variable with firm fixed effects is of the order of the inverse of the number of periods and is therefore small.

4. Empirical Results

4.1 Descriptive Statistics

Table 1 reports the distribution of the variables used in the analysis. The mean quarter-end CDS spread is 160 basis points and the median is 93 basis points. The gap between the two confirms the strong right skew that motivates the logarithmic transformation, since the mean and median of the log spread are 4.63 and 4.53. The quarterly-average spread has an almost identical distribution to the quarter-end spread, which suggests that the choice between the two measures is unlikely to be consequential. The quarterly change in the log spread has a mean of -0.02 , consistent with the general decline in credit spreads over the sample period.

Controversy exposure is frequent but unevenly distributed. The mean number of sharply matched incidents is 3.00 per firm-quarter against a median of one, and no sharp incident is recorded in 38.2 per cent of the firm-quarters. Within the reach classification, medium-reach incidents are the most frequent, with a mean of 1.51 per firm-quarter, while high-reach incidents are the least frequent, with a mean of 0.67. Within the severity classification, low-severity incidents dominate with a mean of 2.25, while high-severity incidents are rare, with a mean of 0.04 and a maximum of three. The limited variation in high-severity incidents restricts the precision with which their effect can be estimated. Of the 98 firms, 57 are classified as speculative grade. Their mean quarter-end spread is 225 basis points, more than twice the mean of 84 basis points recorded for investment grade firms.

Table 1. *Descriptive statistics*

	min	max	mean	median	sd
cds_eoq	12.93	2931.30	160.44	93.13	194.84
cds_quarter_average	12.69	2684.20	159.56	93.19	193.00
ln CDS spread (quarter-end)	2.56	7.98	4.63	4.53	0.91
ln CDS spread (quarterly average)	2.54	7.90	4.63	4.53	0.91
Change in ln CDS spread	-1.62	2.33	-0.02	-0.02	0.25
Controversy incidents (sharp)	0.00	55.00	3.00	1.00	5.44
Controversy incidents (all)	0.00	57.00	3.45	1.00	5.90
Low-reach incidents	0.00	21.00	1.26	0.00	2.08
Medium-reach incidents	0.00	31.00	1.51	0.00	2.79
High-reach incidents	0.00	27.00	0.67	0.00	1.80
Low-severity incidents	0.00	41.00	2.25	1.00	4.04
Medium-severity incidents	0.00	31.00	1.16	0.00	2.22
High-severity incidents	0.00	3.00	0.04	0.00	0.25
Average reach	0.00	3.00	1.15	1.40	0.91
Average severity	0.00	3.00	0.91	1.00	0.71
Current RepRisk Index	0.00	73.00	22.93	22.00	14.10
ESG score (t-1)	7.40	92.21	58.67	62.09	18.55
High yield	0.00	1.00	0.54	1.00	0.50
Leverage	-22.59	39.69	1.65	1.01	5.73
Current ratio	0.32	9.79	1.82	1.39	1.44
Profit margin	-16.78	50.99	11.68	10.23	10.83
ln Sales	12.96	19.26	16.42	16.24	1.31

Figure 1 shows the industry composition of the sample. Consumer cyclical firms form the largest group with 30 firms, followed by consumer non-cyclical firms with 24 firms. The remaining six industry groups together account for 44 firms. The concentration in consumer-related sectors is relevant for interpretation, since these firms are subject to greater public and media attention. It is also one reason for the use of industry-by-quarter fixed effects in the robustness analysis.

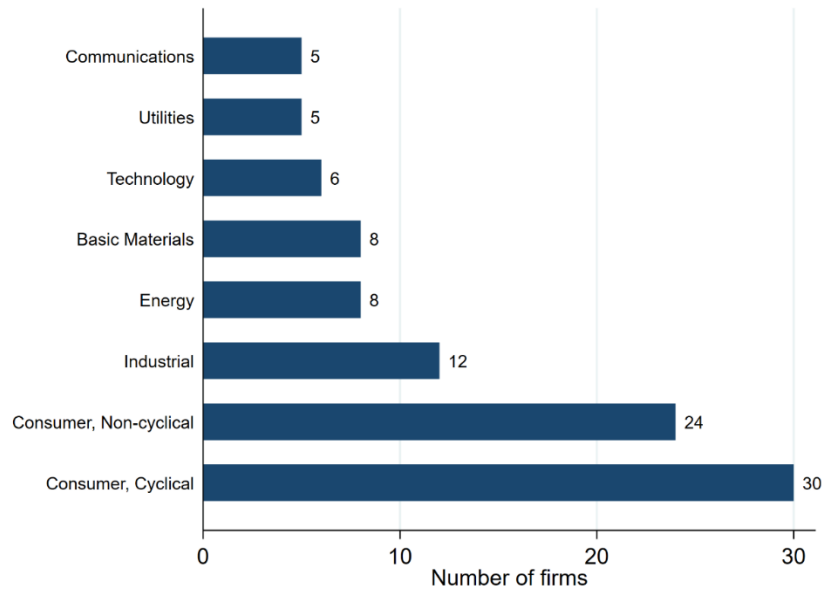


Figure 1. Sample distribution across industries

Figure 2 plots the median quarter-end CDS spread against the mean number of sharply matched incidents per firm. The two series move in opposite directions over the sample period. Credit spreads decline as markets normalise after the financial crisis, while recorded controversy exposure rises as RepRisk expands its source coverage. This divergence provides a first indication that a regression of the level of spreads on incident counts may be confounded by trends that are unrelated to credit risk. It motivates both the inclusion of quarter fixed effects and the change specification introduced in Section 4.4.

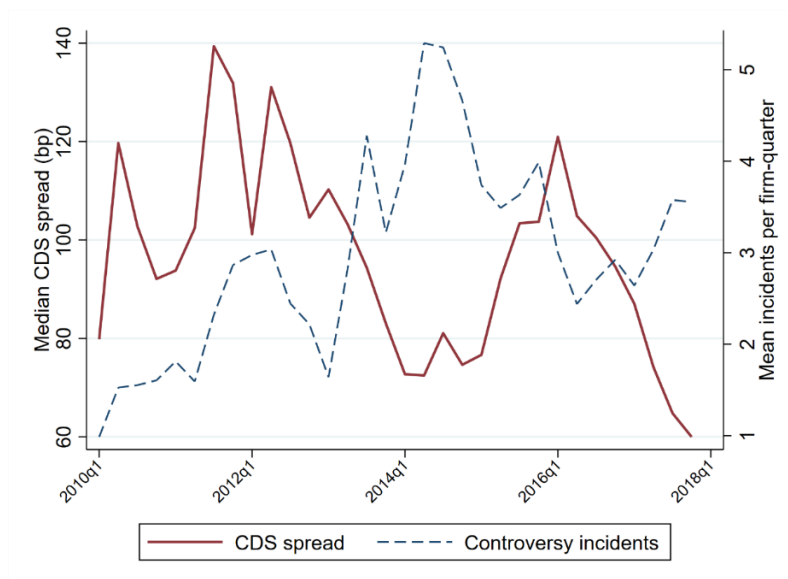


Figure 2. CDS spreads and ESG controversy incidents over time

4.2 Correlations

Table 2 reports Pearson correlations. Both the incident counts and the Current RepRisk Index are negatively correlated with the log CDS spread at the bivariate level. The correlation is -0.28 for *IncidentsSharp* and -0.36 for the Current RepRisk Index, which is contrary to the predicted positive association. The pattern is consistent with a visibility effect rather than a risk effect. Incident counts are positively correlated with firm size at 0.25 and with the ESG score at 0.33 , and negatively correlated with the high yield indicator at -0.08 , while firm size is strongly negatively correlated with the spread at -0.39 . Larger and more closely monitored firms therefore accumulate more recorded incidents while displaying lower credit risk. These bivariate correlations do not control for firm characteristics or common time effects and are not informative about the within-firm relationship that the fixed effects estimator identifies.

With the exception of high-severity incidents (0.42), the reach and severity counts are highly correlated with the total count, with coefficients between 0.83 and 0.95 . This confirms that they cannot be included in the same specification as the total number of incidents. Average reach and average severity are much less strongly

correlated with the count, at 0.43 and 0.34, and can therefore be estimated jointly with it. The correlations among the accounting controls are low and do not indicate multicollinearity concerns.

Table 2. *Correlation matrix*

	In CDS spread (quarter-end)	Controversy incidents (sharp)	Low-reach incidents	Medium-reach incidents	High-reach incidents	Low-severity incidents	Medium-severity incidents	High-severity incidents	Average reach	Average severity	Current RepRisk Index	ESG score (t- 1)	High yield	Leverage	Current ratio	Profit margin
In CDS spread (quarter-end)	1															
Controversy incidents (sharp)	-0.280***	1														
Low-reach incidents	-0.285***	0.845***	1													
Medium-reach incidents	-0.278***	0.932***	0.746***	1												
High-reach incidents	-0.203***	0.829***	0.557***	0.666***	1											
Low-severity incidents	-0.294***	0.955***	0.831***	0.904***	0.785***	1										
Medium-severity incidents	-0.234***	0.849***	0.772***	0.802***	0.705***	0.691***	1									
High-severity incidents	-0.103***	0.423***	0.367***	0.422***	0.321***	0.348***	0.389***	1								
Average reach	-0.300***	0.428***	0.322***	0.432***	0.396***	0.433***	0.364***	0.125***	1							
Average severity	-0.258***	0.336***	0.407***	0.337***	0.219***	0.278***	0.456***	0.177***	0.778***	1						
Current RepRisk Index	-0.357***	0.777***	0.710***	0.751***	0.608***	0.756***	0.693***	0.283***	0.684***	0.608***	1					
ESG score (t-1)	-0.383***	0.331***	0.343***	0.329***	0.227***	0.336***	0.294***	0.119***	0.378***	0.365***	0.451***	1				
High yield	0.425***	-0.0766***	-0.0822***	-0.0702***	-0.0722***	-0.0977***	-0.0481*	0.0179	-0.194***	-0.149***	-0.211***	-0.183***	1			
Leverage	-0.0649***	-0.0443*	-0.0240	-0.0381*	-0.0457*	-0.0352	-0.0401*	-0.0279	-0.0314	-0.0379*	-0.0486*	0.00388	0.0231	1		
Current ratio	0.199***	-0.0546**	-0.0803***	-0.0636***	-0.0254	-0.0666***	-0.0544**	-0.000710	-0.156***	-0.151***	-0.143***	-0.301***	0.110***	-0.0225	1	
Profit margin	-0.163***	0.109***	0.0929***	0.0973***	0.0907***	0.102***	0.0980***	-0.00318	0.160***	0.123***	0.167***	0.139***	-0.0491*	-0.000260	-0.111***	1
In Sales	-0.394***	0.251***	0.233***	0.231***	0.228***	0.277***	0.184***	0.0409*	0.276***	0.216***	0.354***	0.279***	-0.525***	-0.0220	-0.314***	0.0179

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

4.3 Baseline Results

Table 3 reports the baseline regressions of the log quarter-end CDS spread on controversy exposure, with firm and quarter fixed effects and standard errors two-way clustered by firm and quarter. Column 0 contains the controls only. Column 1 adds *IncidentsSharp*. Columns 2 and 3 add average reach and average severity separately, column 4 includes both, and column 5 adds their interaction.

The coefficient on *IncidentsSharp* is negative and statistically significant in every specification, with a value of -0.0126 in column 1. H1 is therefore not supported in the level specification. However, the result should not be read as evidence that controversies reduce credit risk. As the descriptive evidence indicates, the incident count measures both adverse events and the intensity of external monitoring. This partly happens because RepRisk expanded its media coverage over time, so the same firm shows more recorded incidents later on, not because it behaved worse. This coincides with a period when many of these firms were growing and improving in credit quality. Sections 4.4 and 4.5 show that this negative coefficient is a property of the level of spreads only. It halves when firm-specific trends are included and it disappears entirely once the persistence of the spread itself is controlled for.

Average reach entered alone is positive and significant at the 10 per cent level (column 2), but loses significance once average severity is included jointly (column 4). Using the coefficient of 0.0247 from column 2, a one-point increase in average reach on the three-point RepRisk scale is associated with a spread that is approximately 2.5 per cent higher, or roughly 4 basis points at the sample mean. The centred specification in column 5 yields a similar magnitude (0.0357 , significant at the 10 per cent level). H2 therefore receives partial support: the reach effect is significant when tested on its own but weakens once average severity is added to the same specification, which is consistent with the low variation in severity discussed below. Conditional on the number of incidents recorded, controversies disseminated by higher-reach sources are associated with higher credit spreads.

Average severity is economically small and statistically insignificant in every

specification. Its interaction with average reach is positive as predicted but not significant, with a coefficient of 0.0293. H3 is not supported. This is consistent with Gutknecht (2025), who finds that severity does not consistently moderate the relationship between controversy coverage and equity risk. Two explanations are plausible. Severity as classified by RepRisk reflects the nature of the alleged misconduct rather than its financial consequence. In addition, the classification is heavily concentrated in the low category, which limits the variation available for identification.

Among the controls, the current ratio is consistently negative and highly significant, as expected for a measure of short-term solvency. The ESG score carries a negative coefficient throughout but is not significant within firms. This is unsurprising given that ESG scores move slowly and that most of their variation is cross-sectional. The coefficient is negative and significant at the 5 per cent level in the industry fixed effects specification reported in Table A.1.

Table 3. *ESG controversies and the level of CDS spreads*

	(0)	(1)	(2)	(3)	(4)	(5)
Controversy incidents (sharp)		-0.0126*** (0.0032)	-0.0137*** (0.0033)	-0.0128*** (0.0032)	-0.0138*** (0.0033)	-0.0139*** (0.0033)
Average reach			0.0247* (0.0134)		0.0330 (0.0199)	
Average reach (centred)						0.0357* (0.0193)
Average severity				0.0136 (0.0164)	-0.0154 (0.0237)	
Average severity (centred)						-0.0068 (0.0264)
Average reach x severity						0.0293 (0.0204)
ESG score (t-1)	-0.0035 (0.0027)	-0.0035 (0.0026)	-0.0036 (0.0026)	-0.0035 (0.0026)	-0.0036 (0.0026)	-0.0036 (0.0026)
ln Sales	0.0006 (0.1252)	0.0077 (0.1240)	0.0066 (0.1236)	0.0064 (0.1243)	0.0077 (0.1240)	0.0077 (0.1241)
Leverage	-0.0031 (0.0029)	-0.0032 (0.0028)	-0.0032 (0.0029)	-0.0032 (0.0028)	-0.0032 (0.0029)	-0.0032 (0.0029)
Current ratio	-0.0773*** (0.0240)	-0.0807*** (0.0238)	-0.0793*** (0.0236)	-0.0802*** (0.0237)	-0.0794*** (0.0236)	-0.0795*** (0.0236)
Profit margin	0.0009 (0.0034)	0.0007 (0.0032)	0.0006 (0.0032)	0.0007 (0.0032)	0.0006 (0.0032)	0.0006 (0.0032)
Observations	2737	2737	2737	2737	2737	2737
Within R-squared	0.017	0.028	0.030	0.028	0.030	0.031
Overall R-squared	0.851	0.853	0.853	0.853	0.853	0.853

Standard errors in parentheses. Dependent variable: ln CDS spread at quarter end. Firm and quarter fixed effects included in all columns. Standard errors two-way clustered by firm and quarter. Within R-squared is net of firm and quarter fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

4.4 Controversy Exposure and the Repricing of Credit Risk

Table 4 turns from the level of spreads to their movement. Panel A reports regressions of the quarterly change in the log spread on controversy exposure measured in levels. Panel B adds the lagged log spread to the level equation.

The results reverse the sign obtained in Table 3. In Panel A, *IncidentsSharp* enters positively with a coefficient of 0.0020 in column 1, but it is not statistically significant. The number of controversies recorded in a quarter therefore does not by itself predict a widening of the spread. The reach dimension is the decisive one. Average reach is positive and significant at the 5 per cent level, with a coefficient of 0.0143 in column 2. The decomposition in column 3 shows that this effect is driven

exclusively by the most visible incidents. The coefficient on high-reach incidents is 0.0056 and is significant at the 10 per cent level, while low-reach and medium-reach incidents are indistinguishable from zero and the latter is even negative. The Current RepRisk Index, which weights incidents by reach, severity and novelty, is positive but not statistically significant in column 5. Within the severity decomposition in column 4, only medium-severity incidents attract a significant coefficient. Low-severity incidents are indistinguishable from zero, and high-severity incidents remain insignificant with a standard error more than twenty times the size of the coefficient, which reflects the small number of such events in the sample.

Panel B shows why the level and change specifications differ. The lagged log spread enters with a coefficient of 0.824, confirming that credit spreads are highly persistent. Once this persistence is accounted for, the coefficient on *IncidentsSharp* is economically negligible and statistically indistinguishable from zero at -0.0004 , while average reach remains positive and significant at the 5 per cent level at 0.0158. The implied long-run effect of a one-point increase in average reach is 0.090, which corresponds to a spread that is 9 per cent higher. The negative coefficient in Table 3 is therefore not a repricing effect. It reflects persistent cross-sectional differences in the level of spreads between firms that attract more and less media attention, which firm fixed effects do not fully absorb because media attention itself evolves over time. Once the analysis focuses on movements in spreads, the predicted direction is recovered for the reach dimension.

Table 4. *Short-run repricing and dynamic specifications*

Panel A. *Change in ln CDS spread*

	(1)	(2)	(3)	(4)	(5)
Controversy incidents (sharp)	0.0020 (0.0014)	0.0014 (0.0014)			
Average reach		0.0143** (0.0069)			
Low-reach incidents			0.0041 (0.0042)		
Medium-reach incidents			-0.0013 (0.0022)		
High-reach incidents			0.0056* (0.0033)		
Low-severity incidents				-0.0003 (0.0014)	
Medium-severity incidents				0.0071** (0.0028)	
High-severity incidents				0.0008 (0.0205)	
Current RepRisk Index					0.0009 (0.0006)
Observations	2,639	2,639	2,639	2,639	2,639
R-squared	0.312	0.313	0.312	0.313	0.312

Panel B. *Dynamic specification*

	(1)	(2)	(3)
Controversy incidents (sharp)	-0.0004 (0.0015)	-0.0011 (0.0014)	-0.0011 (0.0014)
L.ln CDS spread (quarter-end)	0.8242*** (0.0293)	0.8236*** (0.0291)	0.8239*** (0.0291)
Average reach		0.0158** (0.0067)	
Average reach (centred)			0.0130 (0.0093)
Average severity (centred)			0.0136 (0.0107)
Average reach x severity			0.0180 (0.0158)
Observations	2,639	2,639	2,639
R-squared	0.952	0.952	0.952

Panel A reports the change in ln CDS spread regressed on the level of controversy exposure, isolating within-firm repricing at the time of the event. Panel B reports the level of ln CDS spread with the lagged dependent variable included, isolating the component of the spread that deviates from its own persistence. Controls, firm and quarter fixed effects included in all columns. Standard errors two-way clustered by firm and quarter. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

4.5 Robustness

The robustness analysis addresses four concerns. The first is the measurement of the dependent variable. The quarter-end CDS spread is replaced with the quarterly-average CDS spread, and the change specification is re-estimated using the quarterly average. The second is the measurement of controversy exposure. *IncidentsSharp* is replaced with *IncidentsAll*, with the logarithmic form of each count, and with exposure lagged by one quarter. The third is the possibility that the results reflect differential trends rather than controversy exposure. Firm-specific linear trends are added to the baseline model, and the quarter fixed effects are replaced with industry-by-quarter fixed effects, which absorb time-varying shocks common to an industry, such as the decline in energy prices in 2014 and 2015. The fourth is the specification of fixed effects and inference. Since the primary specification throughout the thesis clusters standard errors by firm and quarter, the baseline is also re-estimated with industry-by-quarter clustering for comparison, given that industry-based clustering is common in the related literature. In addition, the high yield classification does not vary over time in the available data, so its level effect is absorbed by the firm fixed effects. A model with industry fixed effects rather than firm fixed effects is therefore estimated so that the level effect of the credit-rating category can be recovered.

Table A.1 reports the results. The negative coefficient on controversy incidents is present in every level specification and remains significant at the 1 per cent level in columns 1, 2 and 4, at -0.0147 when the quarterly-average spread replaces the quarter-end spread, at -0.0132 when all incidents replace sharply matched incidents, and at -0.0119 when exposure is lagged by one quarter. Replacing the count with its logarithmic transformation in column 3 gives a coefficient of -0.0733 , significant at the 5 per cent level.

Columns 5 and 6 address the divergent trends visible in Figure 2. Adding firm-specific linear trends reduces the coefficient on controversy incidents from -0.0138 in the baseline to -0.0063 , a decline of more than half. This supports the interpretation that a substantial part of the negative level relationship reflects firm-level trends in

media coverage rather than credit risk. Replacing quarter fixed effects with industry-by-quarter fixed effects reduces the coefficient to -0.0088 and strengthens the reach effect to 0.0495 , significant at the 5 per cent level.

The coefficient on average reach is positive in seven of the nine columns, but turns negative, though insignificant, once industry fixed effects replace firm fixed effects in column 7. Its statistical significance varies across specifications: it is significant at the 5 per cent level in the logarithmic specification in column 3 and under industry-by-quarter fixed effects in column 6, at the 5 per cent level when exposure is lagged in column 4, and at the 10 per cent level under industry-by-quarter clustering in column 8. It is positive but insignificant when the quarterly-average spread is used in column 1, when all incidents are used in column 2, and when firm-specific trends are added in column 5.

Column 7 reports the specification with industry rather than firm fixed effects, in which the level effect of the credit-rating category can be estimated. The high yield indicator carries a coefficient of 0.5949 , which implies that speculative-grade firms trade at spreads approximately 81 per cent higher than investment-grade firms with comparable characteristics. The lagged ESG score enters negatively and is significant at the 5 per cent level with a coefficient of -0.0082 . Average reach is not significantly identified in this specification, and enters with a negative sign, since without firm fixed effects the variable is dominated by cross-sectional differences in media exposure.

Column 8 re-estimates the baseline with standard errors clustered by industry and quarter. The coefficients are identical to the baseline by construction; inference is broadly unchanged for controversy incidents, and average reach becomes significant at the 10 per cent level, compared with being insignificant at conventional levels under firm-by-quarter clustering. This illustrates that the small number of industry clusters can produce tighter standard errors than the firm-by-quarter alternative, which is the reason firm-by-quarter clustering is used as the primary specification throughout.

Column 9 reports the change specification with the quarterly-average spread as the dependent variable. The coefficients retain their sign but are smaller and statistically

insignificant, as expected given that the quarterly average is partly measured before the events recorded in the same quarter.

5. Conclusions

This thesis examined whether, and how, ESG controversy events are priced in the market for corporate credit risk, using CDS spreads as a direct market-based measure of that risk. Building on the reach-and-severity mechanism identified by Kölbel et al. (2017), it tested three hypotheses. The first hypothesis, H1, is that controversy events are positively associated with CDS spreads. The second hypothesis, H2, is that this association is stronger for controversies reported by higher-reach media sources. The third hypothesis, H3, is that the severity of the underlying misconduct further amplifies the relationship, conditional on high reach. Using a panel of 98 non-financial firms and 2,737 firm-quarter observations between 2010 and 2017, the analysis tested these hypotheses in a level specification, a change specification, and a dynamic specification that accounts for the persistence of CDS spreads.

The results do not support H1 in its simplest form. The number of controversy incidents recorded for a firm in a quarter is negatively associated with the level of its CDS spread, not positively. This finding is unlikely to mean that controversies actually reduce credit risk. A more plausible explanation is that incident counts recorded in media-based databases such as RepRisk capture two things at once. They capture the occurrence of adverse events, and they also capture how closely a firm is monitored by the media. RepRisk expanded its media coverage over the sample period, so a given firm accumulates more recorded incidents later in the sample even without any change in its underlying behavior. This coincided with a period in which the credit quality of many sample firms was improving for unrelated reasons. Firm fixed effects cannot absorb this confound, because it reflects a trend shared across firms over time rather than a fixed difference between firms. Once the analysis moves from the level of spreads to their change, and once the persistence of spreads is modelled explicitly, the coefficient on incident count becomes statistically indistinguishable from zero rather

than negative. This pattern supports the measurement-related explanation, though it does not prove it. The panel is not long enough, and no instrument for media coverage is available, to rule out other explanations with full certainty.

H2 receives partial support. Average reach is positively associated with CDS spreads in the level specification, and it remains significant in both the change specification and the dynamic specification. The dynamic specification implies a long-run effect of approximately 9 per cent for a one-point increase in reach on RepRisk's three-point scale. This long-run figure comes from a partial-adjustment model and should be read as indicative rather than precisely estimated, because dynamic panel models with fixed effects are known to carry a small-sample bias. The effect appears concentrated in high-reach incidents. These carry a significant coefficient, while low-reach and medium-reach incidents do not. This thesis reports this pattern rather than a formal test of the difference between the coefficients, and such a test would be needed to state that the high-reach effect is significantly larger.

H3 is not supported. The severity of criticism is economically small and statistically insignificant throughout the main analysis. Its interaction with reach is positive, as predicted, but does not reach conventional significance. In one robustness specification that replaces quarter fixed effects with industry-by-quarter fixed effects, while still including firm fixed effects, average severity is significantly negative, which is the opposite of the predicted direction.

Overall, these findings do not identify a previously unstudied relationship between ESG controversies and credit risk. That relationship, and the role of reach and severity within it, was established by Kölbel et al. (2017). The contribution of this thesis is twofold. Methodologically, it shows that the sign and significance of a standard count-based exposure measure depend materially on whether the specification isolates within-firm repricing from persistent cross-sectional differences in media exposure, a distinction likely to matter whenever controversy or incident counts are built from media-based sources. Empirically, it provides an out-of-sample replication of the reach-driven mechanism that Kölbel et al. (2017) document, showing that the effect survives

in a later sample period once this specification issue is addressed.

These findings should be read in light of several limitations. The sample is restricted to 98 non-financial firms with both CDS and RepRisk coverage. This reflects a complete-case selection process that is likely to favour larger, more closely monitored firms. The results should not be extrapolated to smaller or unrated firms.

The panel is unbalanced. The theoretical maximum is 3,136 firm-quarter observations, based on 98 firms over 32 quarters, but the final sample contains only 2,737. This shortfall comes from gaps in CDS quote availability, gaps in lagged accounting and ESG data, and gaps in RepRisk's metrics coverage. The median number of quarters per firm is 32, but the mean is lower. The panel should therefore be described as unbalanced rather than close to balanced.

The analysis matches accounting and ESG data from the prior fiscal year to each quarter. This does not guarantee that the information was actually available to market participants at the start of every quarter. A stricter lag based on actual reporting dates was not implemented.

Average reach and average severity are coded as zero in quarters with no recorded incident. This means their coefficients combine two effects: the effect of a controversy occurring, and the effect of its character once it has occurred. A two-part specification that separates these two effects was not estimated.

The dynamic specification combines firm and quarter fixed effects with a lagged dependent variable. This combination is known to introduce a finite-sample bias. Given 32 quarters per firm, this bias is likely to be modest, but a bias-corrected estimator was not used. The implied long-run effect should be read with this caveat in mind.

Finally, the analysis establishes association within the same quarter rather than a specific sequence of events. The change and dynamic specifications help address concerns about persistent, unrelated differences in media exposure. However, a formal test of whether credit deterioration precedes or follows media coverage was not conducted. Such a test would require lead-lag or event-window analysis.

Future research could address these limitations directly. A larger and more

geographically diverse sample would support finer industry classification, country-level controls, and less constrained inference. Future research could also use a more detailed measure of severity. RepRisk's three-point scale may be too coarse to capture real differences in how serious a controversy is, which could help explain why this thesis finds little effect of severity and even an unexpected negative result in one robustness test. Formal equality tests across the reach and severity categories, together with a bias-corrected estimator for the dynamic specification, would allow the current results to be stated with more confidence. Lead-lag or event-study designs using higher-frequency data would help establish whether credit markets anticipate, coincide with, or lag media coverage of controversies.

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Appendix A

Table A.1. *Robustness tests*

	(1) AvgCDS	(2) AllInc	(3) LogCount	(4) Lagged	(5) FirmTrend	(6) IndxQtr	(7) IndFE	(8) IndClu	(9) D.AvgCDS
Controversy incidents (sharp)	-0.0147*** (0.0035)				-0.0063** (0.0025)	-0.0088*** (0.0029)	-0.0226* (0.0117)	-0.0138*** (0.0021)	0.0015 (0.0012)
Average reach	0.0283 (0.0188)	0.0312 (0.0199)	0.0465** (0.0209)		0.0105 (0.0108)	0.0495** (0.0196)	-0.0600 (0.0462)	0.0330* (0.0166)	0.0043 (0.0056)
Average severity	-0.0195 (0.0222)	-0.0124 (0.0237)	-0.0073 (0.0250)		0.0033 (0.0145)	-0.0504** (0.0205)	0.0308 (0.0390)	-0.0154 (0.0204)	
ESG score (t-1)	-0.0035 (0.0027)	-0.0037 (0.0026)	-0.0036 (0.0026)	-0.0036 (0.0026)	0.0044* (0.0022)	-0.0030 (0.0023)	-0.0082** (0.0031)	-0.0036 (0.0024)	-0.0005 (0.0006)
Controversy incidents (all)		-0.0132*** (0.0033)							
ln(1+incidents, sharp)			-0.0733** (0.0291)						
L.Controversy incidents (sharp)				-0.0119*** (0.0036)					
L.Average reach				0.0232** (0.0109)					
High yield							0.5949*** (0.1676)		
Observations	2737	2737	2737	2639	2737	2737	2737	2737	2639
R-squared	0.858	0.853	0.852	0.856	0.913	0.890	0.430	0.853	0.269

Standard errors in parentheses. Controls, firm fixed effects and quarter fixed effects included in all columns. Column 8 reports standard errors clustered by industry and quarter for comparison; all other columns use firm-by-quarter clustering, the primary specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$